

Linear differential equations of arbitrary order (continuation)

Given: $I \subset \mathbb{R}$ open interval, $a_1, a_2, \dots, a_m \in C(I)$, $f \in C(I)$, $n \in \mathbb{N}^*$

Find $x : I \rightarrow \mathbb{R}$ s.t.

$$(1) \quad x^{(n)} + a_1(t)x^{(n-1)} + \dots + a_{m-1}(t)x' + a_m(t)x = f(t) \quad , \quad t \in \mathbb{R}$$

Remark. n is the order of eq. (1).

$$\underline{L1}: \quad \mathcal{L} : C^n(I) \rightarrow C(I) \quad \mathcal{L}(x)(t) \stackrel{\text{not}}{=} x^{(n)} + a_1(t)x^{(n-1)} + \dots + a_m(t)x, \quad \forall t \in I, \forall x \in C^n(I)$$

$$x \mapsto \mathcal{L}(x)$$

$$(1) \Leftrightarrow \mathcal{L}(x) = f$$

$f \neq 0$ $\mathcal{L}(x) = f$ is said to be a linear non-hom. d.e. (LNHDE)

$f = 0$ $\mathcal{L}(x) = 0$... linear hom. d.e. (LHDE)

we proved that $\mathcal{L}$ is a linear map between the vector spaces $C^n(I)$ and $C(I)$

1. Fundamental theorems for LDE's
2. LHDE's with constant coefficients

1. Fundamental theorems for LDE's

Def. Given $\eta = (\eta_1, \eta_2, \dots, \eta_n) \in \mathbb{R}^n$ and $t_0 \in I$. The following problem is called Initial Value Problem (IVP) or Cauchy Problem (CP):

$$(2) \quad \left\{ \begin{array}{l} \mathcal{L}(x) = f \quad (n^{\text{th}} \text{ order LDE}) \\ x(t_0) = \eta_1 \\ x'(t_0) = \eta_2 \\ \vdots \\ x^{(n-1)}(t_0) = \eta_n \end{array} \right\} \quad n \text{ initial conditions}$$

$t_0 \sim$ the initial time

Example. $n=1$ $\begin{cases} x' + a_1(t)x = f(t) \\ x(t_0) = \eta_1 \end{cases}$ $n=2$ $\begin{cases} x'' + a_1(t)x' + a_2(t)x = f(t) \\ x(t_0) = \eta_1 \\ x'(t_0) = \eta_2 \end{cases}$

Theorem 1 (The existence and uniqueness theorem for the sol. of an IVP)

In the above hypotheses the IVP (2) has a unique solution $\varphi \in C^n(I)$.

Example. Solve the IVP:

$n=1$ $f=0$

$n=2$ $f=0$

Example. Solve the IVP:

$$(i) \begin{cases} x'' + \frac{1}{2}x = 0 \\ x(0) = 0 \\ x'(0) = 0 \end{cases} \quad \begin{matrix} \text{(second order} \\ \text{LHDE)} \\ \text{2 initial} \\ \text{cond.} \end{matrix}$$

First we notice that $x=0$ is a solution.
Then we apply the F! th. and deduce that there are no other solutions.
Hence, $x=0$ is the unique sol.

Theorem 2 (The fundamental theorem for LHDE's)

$\text{Ker } L$ is a vector space of dimension n .

Hence, if $x_1, x_2, \dots, x_n$ are linearly independent solutions of $L(x)=0$, then the general solution of $L(x)=0$ is

$$x = c_1 x_1 + c_2 x_2 + \dots + c_n x_n, \quad c_1, \dots, c_n \in \mathbb{R} \text{ arbitrary.}$$

Examples (i) Prove that $x = c e^{-\frac{t}{2}}$, $c \in \mathbb{R}$ is the general sol. of $x' + \frac{1}{2}x = 0$
(Recall that in Lecture 1 we already proved this by integrating this DE).

Note that $x_1 = e^{-\frac{t}{2}}$ satisfies $x_1 \neq 0$ and x_1 is a sol. of $x' + \frac{1}{2}x = 0$. (Lecture 1)

Th2 $\Rightarrow$ the gen. sol. is $x = c \cdot x_1$, $c \in \mathbb{R}$, i.e. $x = c e^{-\frac{t}{2}}$, $c \in \mathbb{R}$.

(ii) Prove that $x = c_1 e^t + c_2 e^{-t}$, $c_1, c_2 \in \mathbb{R}$ is the general sol. of $x'' - x = 0$
denote $x_1 = e^t$ and $x_2 = e^{-t}$.

we immediately check that x_1 and x_2 are solutions of $x'' - x = 0$;

and that they are lin. independent: take $\alpha_1, \alpha_2 \in \mathbb{R}$ s.t. $\alpha_1 e^t + \alpha_2 e^{-t} = 0$
 $\forall t \in \mathbb{R}$

HW: prove that $\alpha_1 = \alpha_2 = 0$.

Then, by Th2 we get the conclusion

(iii) Prove that $x = c_1 \cosh t + c_2 \sinh t$, $c_1, c_2 \in \mathbb{R}$
is the general sol. of $x'' - x = 0$.

Notation:

$$\begin{cases} \cosh t = \frac{1}{2}(e^t + e^{-t}) & \text{and} & \sinh t = \frac{1}{2}(e^t - e^{-t}) \\ \cosh 0 = 1, \sinh 0 = 0 & (\cosh t)' = \sinh t, (\sinh t)' = \cosh t \\ \cosh^2 t - \sinh^2 t = 1 & \text{HW: represent their graph.} \end{cases}$$

$$\begin{cases} x = \cosh t \\ y = \sinh t, t \in \mathbb{R} \end{cases} \quad x^2 - y^2 = 1 \quad \text{hyperbola} \quad \left| \quad \begin{cases} x = \cos t \\ y = \sin t, t \in \mathbb{R} \end{cases} \quad \cos^2 t + \sin^2 t = 1 \quad \text{circle} \right.$$

$$\left\{ \begin{array}{l} x = \cosh t \\ y = \sinh t, t \in \mathbb{R} \end{array} \right. \quad x^2 - y^2 = 1 \text{ hyperbola} \quad \left| \quad \left\{ \begin{array}{l} x = \cos t \\ y = \sin t, t \in \mathbb{R} \end{array} \right. \quad \cos^2 t + \sin^2 t = 1 \quad x^2 + y^2 = 1 \text{ circle} \right.$$

(represent this hyperbola)

There are two methods: Method 1 Using that $x = c_1 e^t + c_2 e^{-t}$, $c_1, c_2 \in \mathbb{R}$ is the general sol. of $x'' - x = 0$. (HW)

Method 2. Prove first that $x_1 = \cosh t$ and $x_2 = \sinh t$ are linearly indep. solutions of $x'' - x = 0$. The conclusion comes from Th 2. (HW)

Theorem 3 (The fundamental theorem for L.HDE's)

The set of solutions of $L(x) = f$ is $\text{Ker } L + \{x_p\}$, where x_p is a particular solution of $L(x) = f$.

Hence, the general sol. of $L(x) = f$ can be written as $x = x_h + x_p$ where x_h is the general sol. of $L(x) = 0$ (called the L.HDE associated to $L(x) = f$) and x_p is a particular sol. of $L(x) = f$.

Example. Find the general sol. of $x'' - x = -3$.

Step 1 we write the L.HDE associated $x'' - x = 0$ and we find its general sol. we know from a previous example that $x_h = c_1 e^t + c_2 e^{-t}$, $c_1, c_2 \in \mathbb{R}$

Step 2. we look for a partic. sol. of $x'' - x = -3$.

Note that $x_p = 3$.

Step 3 the gen sol is $x = c_1 e^t + c_2 e^{-t} + 3$, $c_1, c_2 \in \mathbb{R}$.

2. The characteristic equation method for L.HDE's with CC (constant coefficients)

$$(3) \quad x^{(m)} + a_1 x^{(m-1)} + \dots + a_{m-1} x' + a_m x = 0 \quad \text{where } a_1, a_2, \dots, a_m \in \mathbb{R}.$$

Idea: Look for solutions of the form $x = e^{rt}$.

$$x' = r e^{rt}, \quad x'' = r^2 e^{rt}, \quad \dots, \quad x^{(k)} = r^k e^{rt}$$

we replace in (3) and obtain: $r^m e^{rt} + a_1 r^{m-1} e^{rt} + \dots + a_{m-1} r e^{rt} + a_m e^{rt} = 0, \forall t \in \mathbb{R}$

$$(4) \quad \underline{r^m + a_1 r^{m-1} + \dots + a_{m-1} r + a_m = 0} \quad \text{this is a polynomial eq. of degree } m \text{ with the same coeff.}$$

as (3)

(4) is called the characteristic equation of the DE (3).

Recall that: (4) has n roots in $\mathbb{C}$, counting with the corresp. multiplicities.

denote the roots by $r_1, r_2, \dots, r_m \in \mathbb{C}$ (some of them are repeated)

e^{rt} when $r \in \mathbb{R}$ of course e^{rt} is a solution of (3) ✓

when $r \in \mathbb{C}$ what e^{rt} means? ?

if $r = -2$ has multiplicity $m = 3$ we have only one sol. e^{-2t} . From where we take other 2 solutions of (3)? ?

There is a particular case in which this idea is sufficient to find the general sol. of (3): when the roots of the ch. eq (4) are real and distinct

The characteristic equation method to find the general sol. of (3) in any situation

Step 1 we write the ch. eq. (4).

Step 2. we find the roots $r_1, r_2, \dots, r_n \in \mathbb{C}$ and their multiplicities (of (4))

Step 3. we associate n functions using the rules:

if $r \in \mathbb{R}$ is a root of (4) of multiplicity $m \geq 1 \mapsto$

$$e^{rt}, te^{rt}, t^2 e^{rt}, \dots, t^{m-1} e^{rt}$$

if $r = \alpha \pm i\beta$ with $\alpha, \beta \in \mathbb{R}, \beta > 0$ are roots of (4) of multiplicity $m \geq 1 \mapsto$

$$e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t, te^{\alpha t} \cos \beta t, te^{\alpha t} \sin \beta t, \dots, t^{m-1} e^{\alpha t} \sin \beta t$$

Step 4. we use the following proposition

Proposition The n functions found at Step 3 are n linearly independent solutions of eq (3). Denote them by $x_1, x_2, \dots, x_n$.

and Th 2 to deduce that the general solution of (3) is

$$x = c_1 x_1 + c_2 x_2 + \dots + c_n x_n, \quad c_1, c_2, \dots, c_n \in \mathbb{R} \text{ arbitrary.}$$

For the next lecture (optional)

- present something about $\csc$, $\sec$
- Proof of Th2 (using Th1) : this is a nice application of linear algebra
- Proof of Proposition above in the case that the roots are real and distinct
- other questions formulated by you.

Euler's formula $e^{ip} = ?$ $e^z, z \in \mathbb{C} ?$